JAVA PROGRAMMING PRACTICES

Farm Animals Using Classes and Objects

Basic Information

Week	Week 4–5
Source file	FarmAnimals.java
Concepts	Classes, objects, constructors, fields, methods.

Objective / Aim

To implement farm animals using classes and objects using the concepts listed in the R23 Java Programming Practices syllabus.

System Requirements

JDK (Java Development Kit) and a text editor or Java IDE.

Algorithm / Logic Description

1. Define an Animal class with details.
2. Create cow, pig, and horse objects.
3. Display each object's details.

UML / Program Structure

```
Animal
+ display()
FarmAnimals creates Cow, Pig, Horse objects
```

Source Code

```
class Animal
{
    String name, habitat, food, sound;
    Animal(String name, String habitat, String food, String sound)
    {
        this.name = name;
        this.habitat = habitat;
        this.food = food;
        this.sound = sound;
    }
    void display()
    {
        System.out.println("Animal: " + name);
        System.out.println("Lives: " + habitat);
        System.out.println("Food: " + food);
        System.out.println("Sound: " + sound + "\n");
    }
}
public class FarmAnimals
{
    public static void main(String[] args)
    {
        Animal cow = new Animal("Cow", "Cowshed", "Grass", "Moo");
        Animal pig = new Animal("Pig", "Pigpen", "Plants and grains", "Oink");
        Animal horse = new Animal("Horse", "Stable", "Grass and hay", "Neigh");
        cow.display();
        pig.display();
        horse.display();
    }
}
```

Compilation and Execution

```
javac FarmAnimals.java
java FarmAnimals
```

Output Screen

```
Animal: Cow
Lives: Cowshed
Food: Grass
Sound: Moo

Animal: Pig
Lives: Pigpen
Food: Plants and grains
Sound: Oink

Animal: Horse
Lives: Stable
Food: Grass and hay
Sound: Neigh
```

Result

The program demonstrates the listed syllabus concepts. The sample output is based on a compiled execution.